

End Semester Examination

July-Aug 2023

EEE2005B - Data Structures and Algorithms

Schedule ID: 20408

Faculty/School	Faculty of Engineering & Technology	Term	IV
Program	Second Year B. Tech	Duration	1 Hours 30 Minutes
Specialization		Max. Marks	40

Read the instructions provided for every question properly before attempting the answer.

Section - 1 : contain(s) **10** question(s) and each question carries **5** mark(s). You can answer any **8** questions out of **10**.

Click **Finish** only after completion of the Exam.

Section - 1 (8 X 5 Marks)

Answer any 8 questions

1	What is data structure? Explain its classification with diagram.	5 marks	CO1	Remembering
2	Find the frequency count for the following: (Note: Steps carries marks) for(i=1;i<=n;i++) for(j=1;j<=n;j++) x=x+1;	5 marks	CO1	Understanding
3	What is SLL? List its operations and explain any one operation with code.	5 marks	CO2	Understanding
4	Perform the following operations on stack, show the representation of stack for each operation and the final content of stack: push(10),push(20),push(30),pop(),push(40).(Note:- Representation Steps carries marks)	5 marks	CO2	Applying
5	Write a C program to implement the conversion of infix expression to postfix expression using stack.	5 marks	CO3	Understanding
6	Explain Circular Queue and its operations with example.(Note: Pseudo code carries marks)	5 marks	CO3	Understanding
7	What is BST property? Explain BT Traversals with example.	5 marks	CO4	Applying
8	Explain the applications of tree.	5 marks	CO4	Understanding

9	Give two differences between trees and graphs. Define graph, adjacency vertex, adjacency edges with example.	5 marks	CO3	Applying
10	Find the Minimum Spanning tree for the following Graph using Kruskal's Algorithm.	5 marks	CO4	Understanding

END OF QUESTION PAPER